

Executive Summary

PASS RATE 34% target 90%	TOTAL TESTS 168 executed	PASSED 58 35% of total	FAILED 110 65% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 34% | Target: 90% | Gap: **+56%**

The Revenue Management System Kenya backend API testing revealed significant reliability concerns with a 34% pass rate across 168 automated tests, substantially below the 90% target threshold. The API Automation suite showed particularly low stability at 33.75% pass rate (27/80), while Backend Smoke Validation performed marginally better at 35.23% (31/88), indicating widespread endpoint instability and response inconsistencies across the system. The high failure rate suggests systemic issues in the backend implementation that require immediate attention to ensure reliable revenue management operations, with failures distributed across both core API functionality and basic smoke validation scenarios.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	27	49	36%	FAIL
2	BE Smoke Validation	88	31	57	35%	FAIL
TOTAL		168	58	110	34%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#42	02 Jul	168	34%	110	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	58	34.5%	180ms	Within SLA
Requests Failed	110	65.5%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	4%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate triage of the 106 failing endpoints to identify common failure patterns and root causes in backend services	Immediate	P1
2	Prioritize stabilization of the 53 API automation failures focusing on response validation and data consistency issues	3 days	P1
3	Review backend service logs and monitoring data to identify performance bottlenecks or timeout issues affecting test execution	5 days	P2
4	Implement enhanced error handling and response validation in the backend API layer to ensure consistent behavior	1 week	P2
5	Deploy incremental fixes starting with smoke validation failures to establish baseline backend stability	1 week	P2
6	Consider implementing backend service health checks and circuit breakers to improve overall system resilience	2 weeks	P3

KEY FINDINGS

1	Backend API endpoints demonstrating 66% failure rate indicating critical stability issues across the revenue management system
2	API Automation suite experiencing 53 endpoint failures suggesting widespread response format or data consistency problems
3	Smoke validation failures in 57 cases pointing to fundamental backend service availability or configuration issues
4	Pass rate differential of 56 percentage points from target indicates backend requires comprehensive stabilization
5	Both test suites showing similar failure patterns suggests systemic backend architectural or implementation challenges

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation		36%
BE Smoke Validation		35%